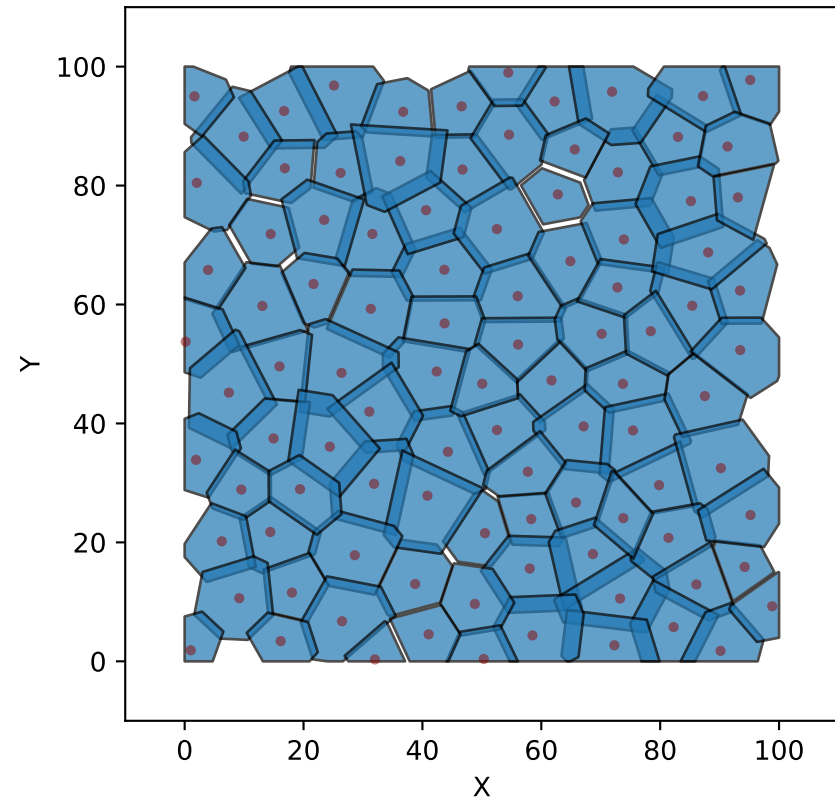
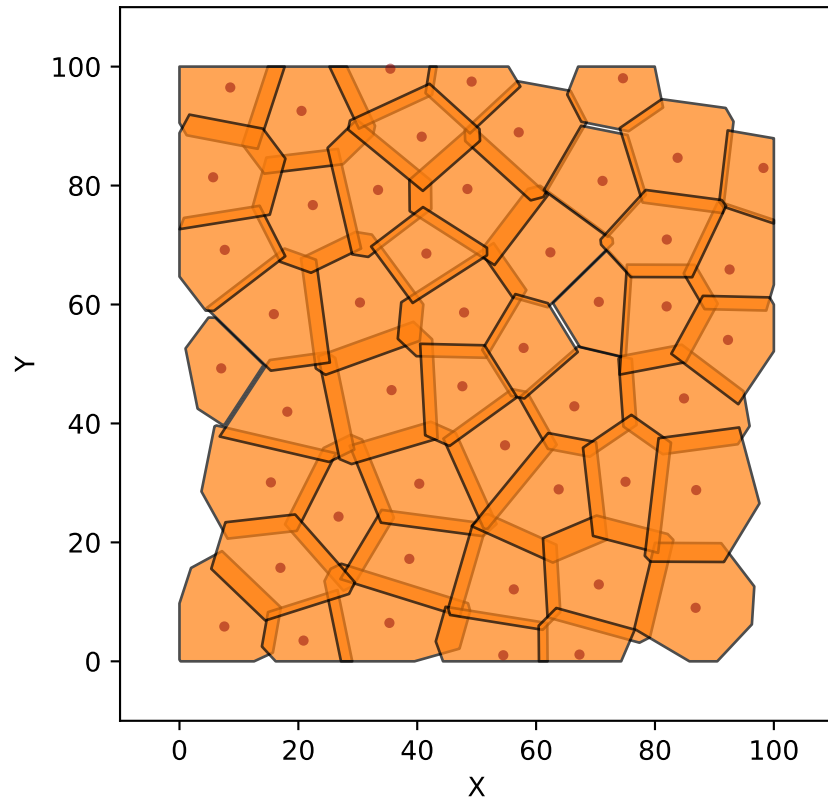
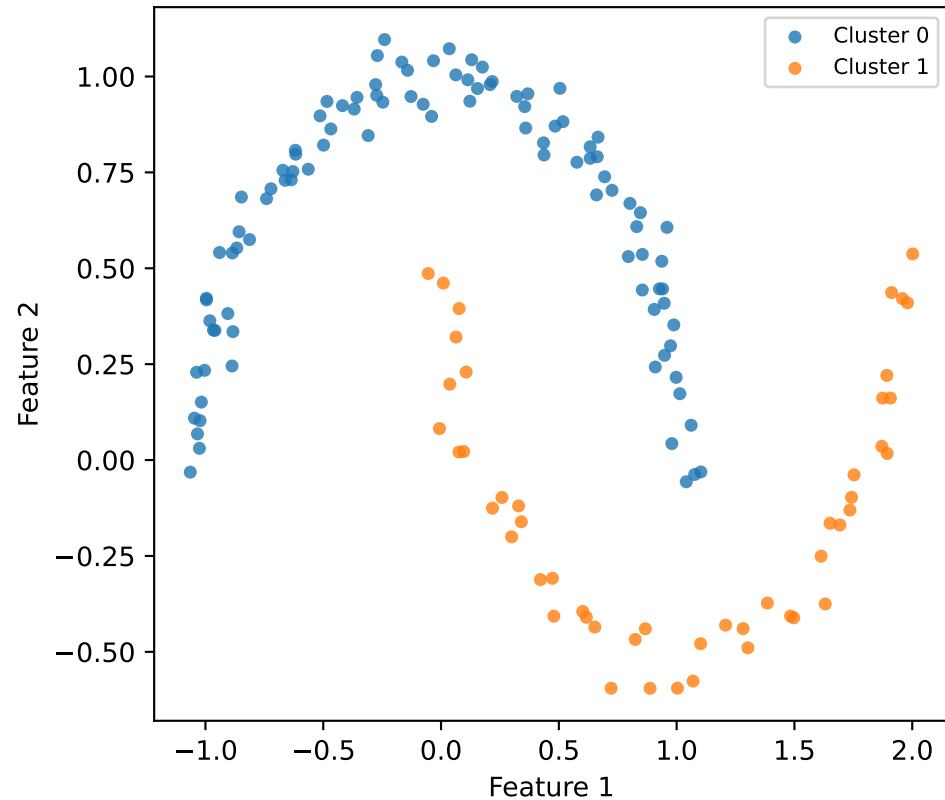


moons — ground truth

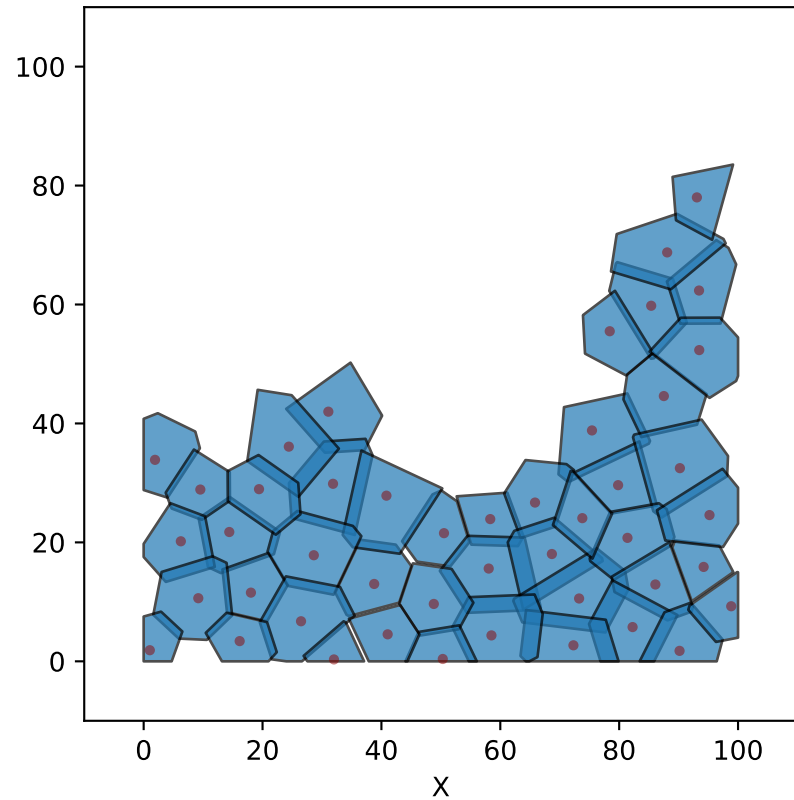
Cluster 0
99 cellsCluster 1
49 cells

Feature space

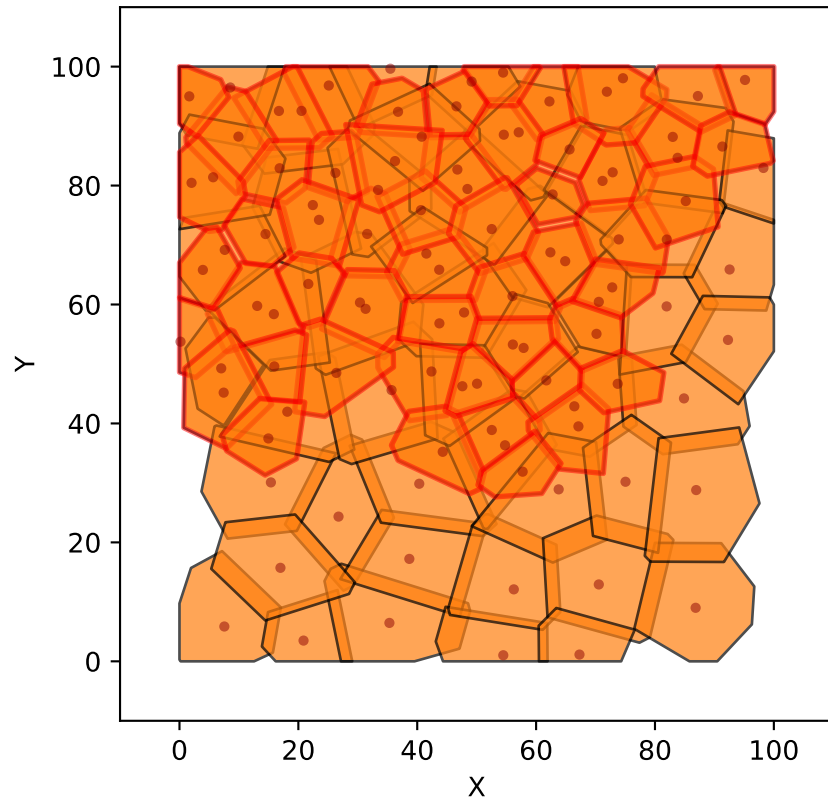


moons — GMM + MRF ARI=0.060 violations 173 -> 170 changed=1 iters=2

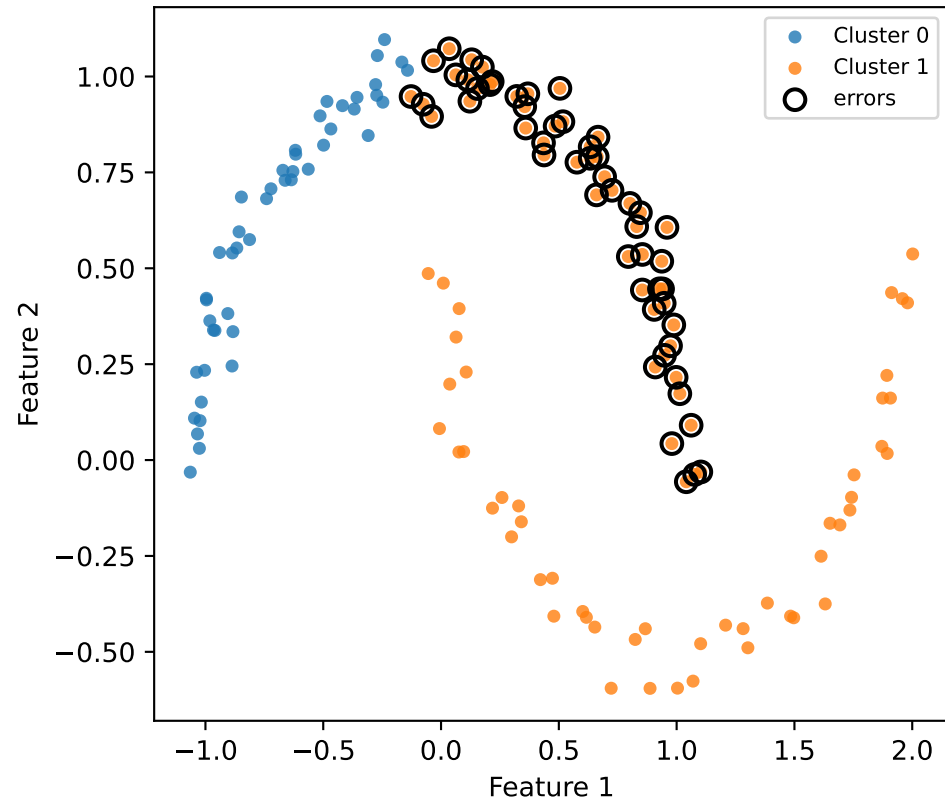
Cluster 0
46 cells



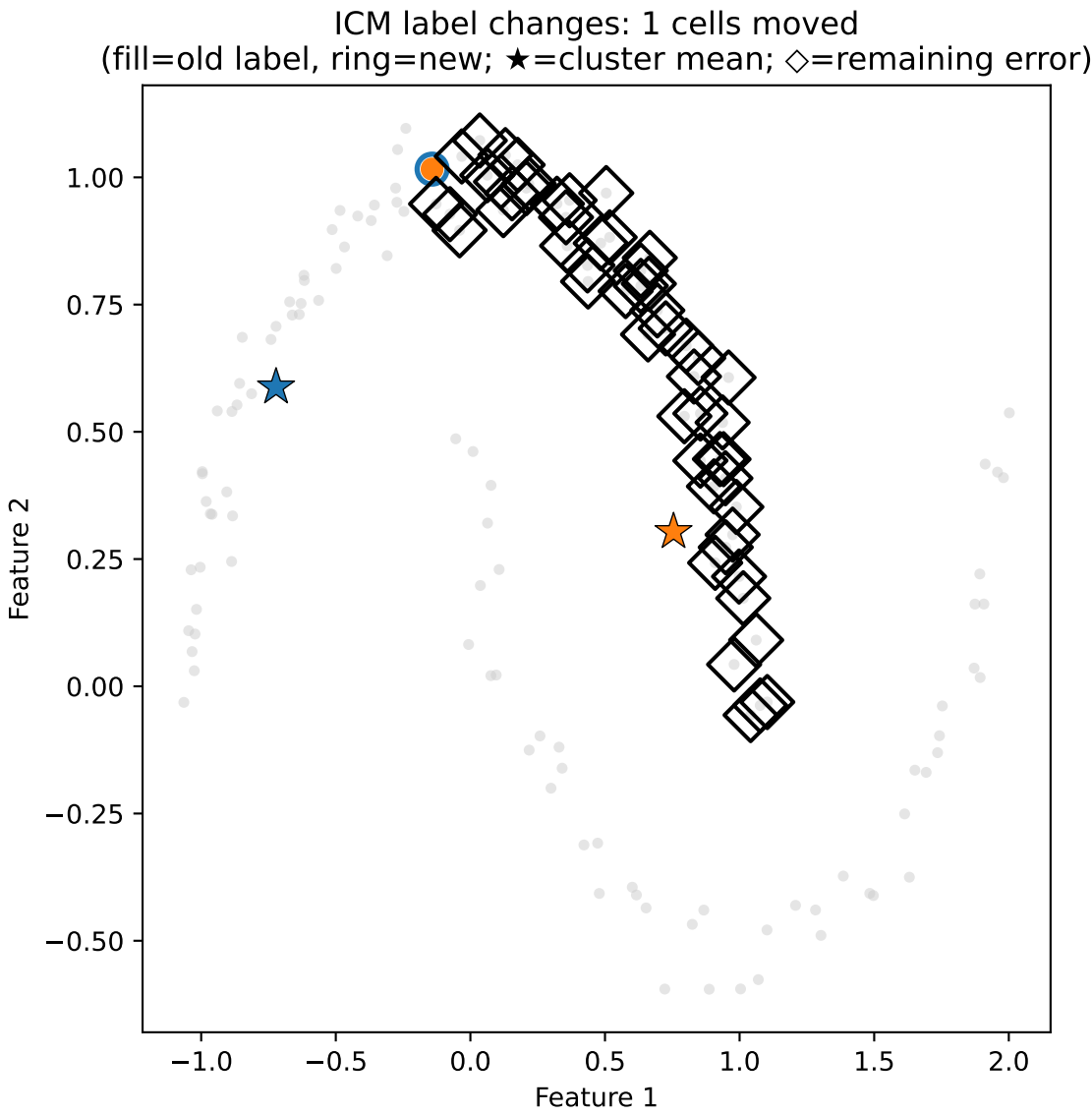
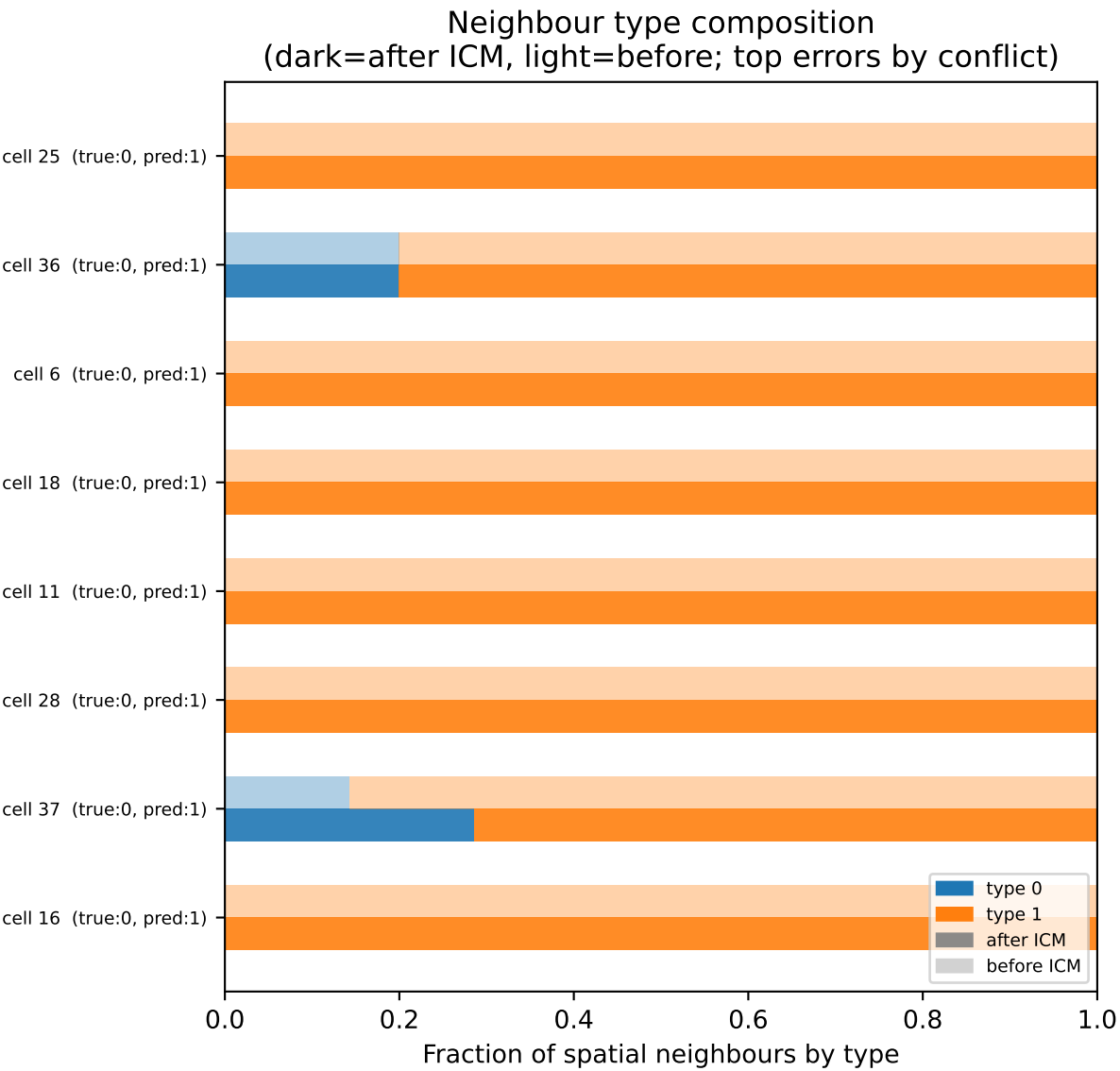
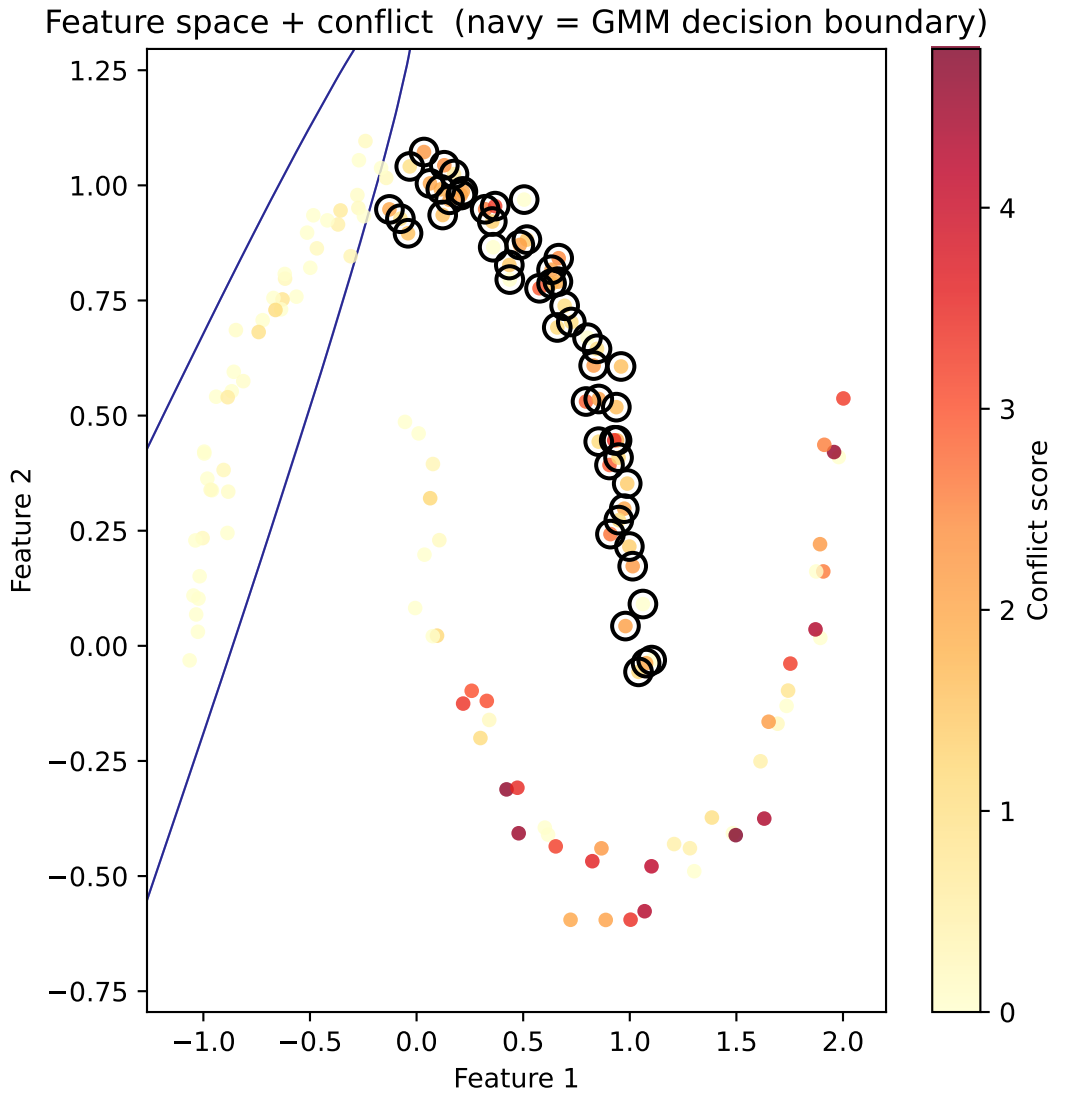
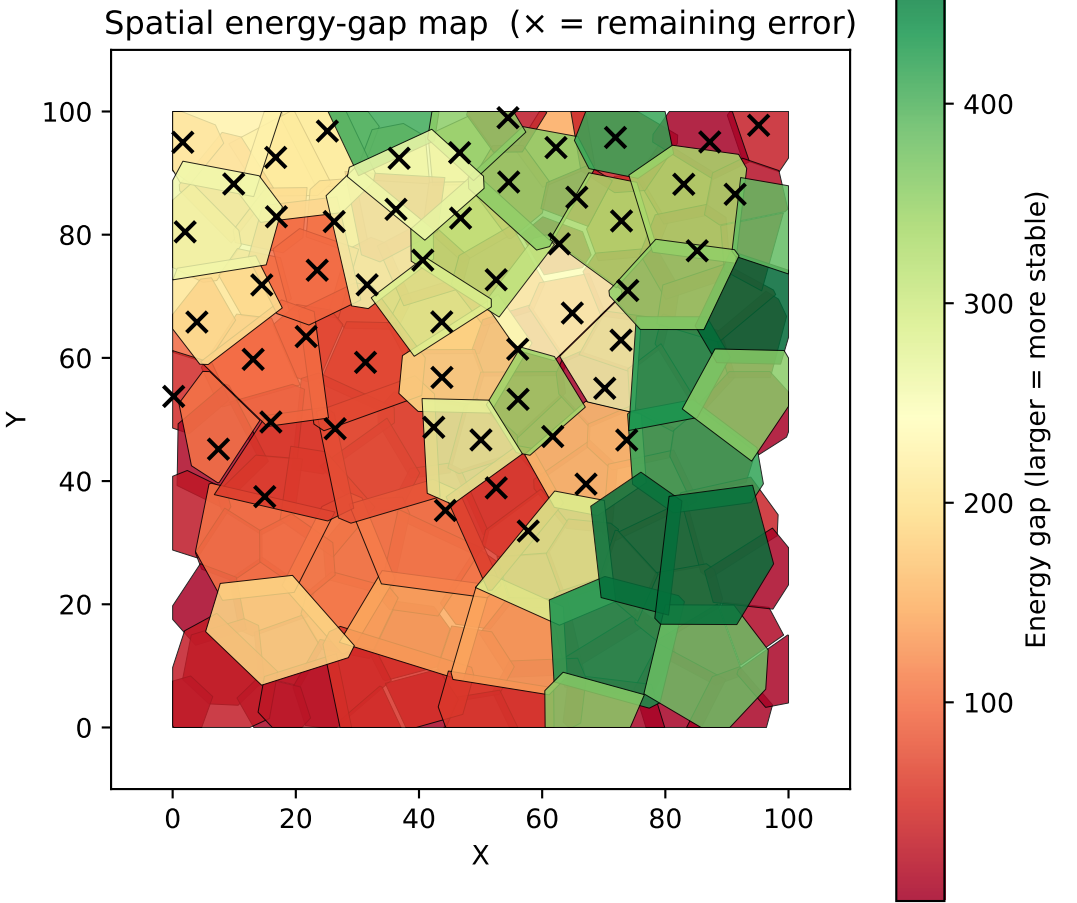
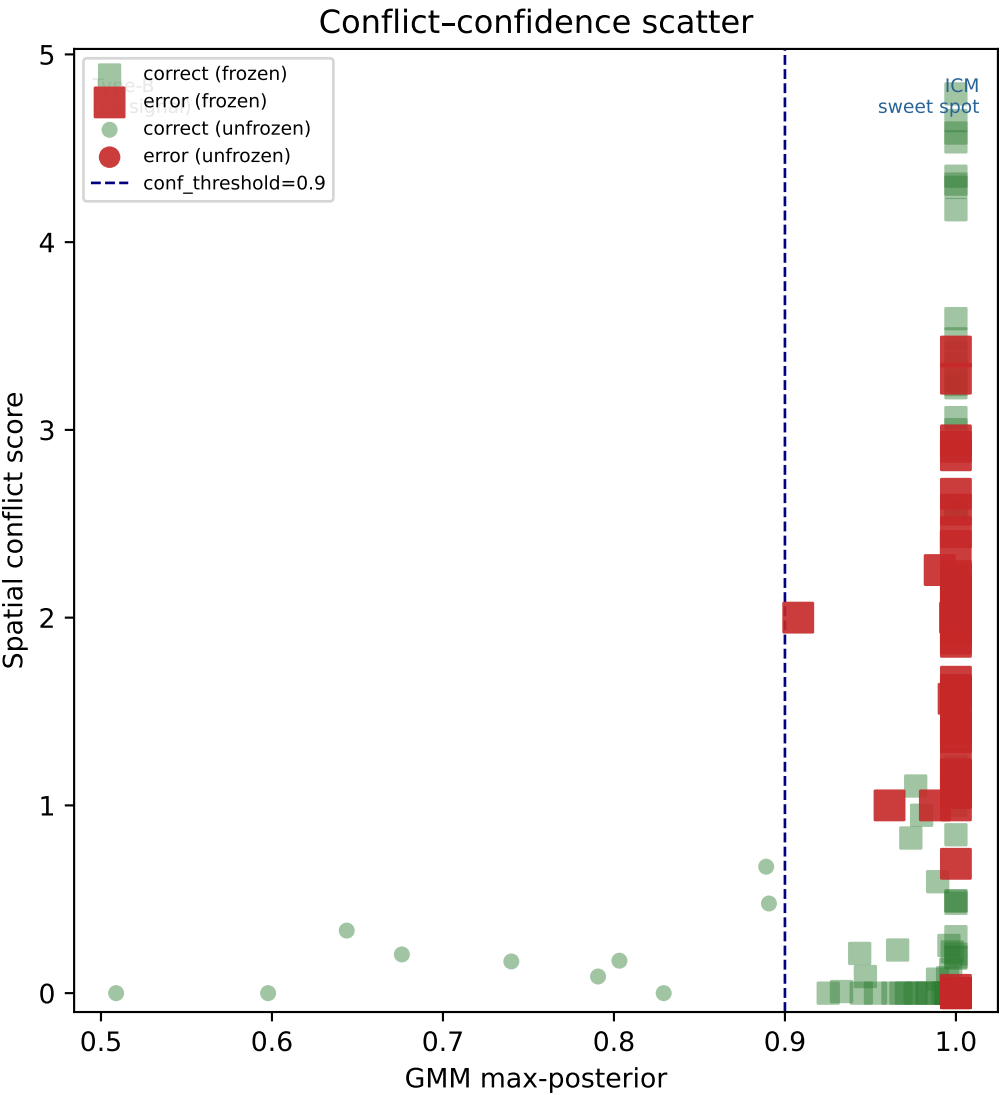
Cluster 1
102 cells (53 err)



Feature space



moons — MRF diagnostics

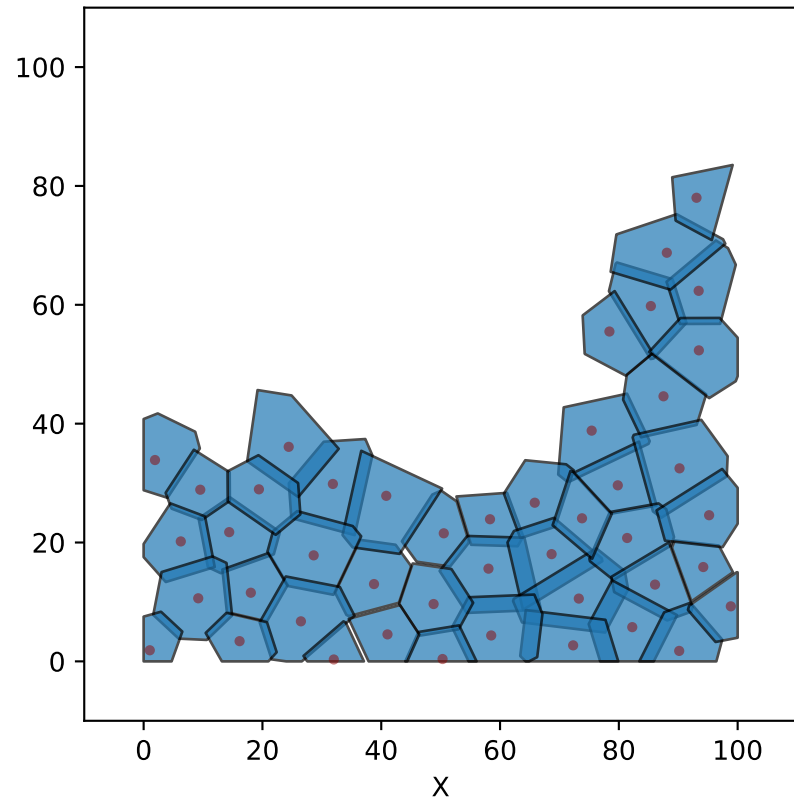


Run summary

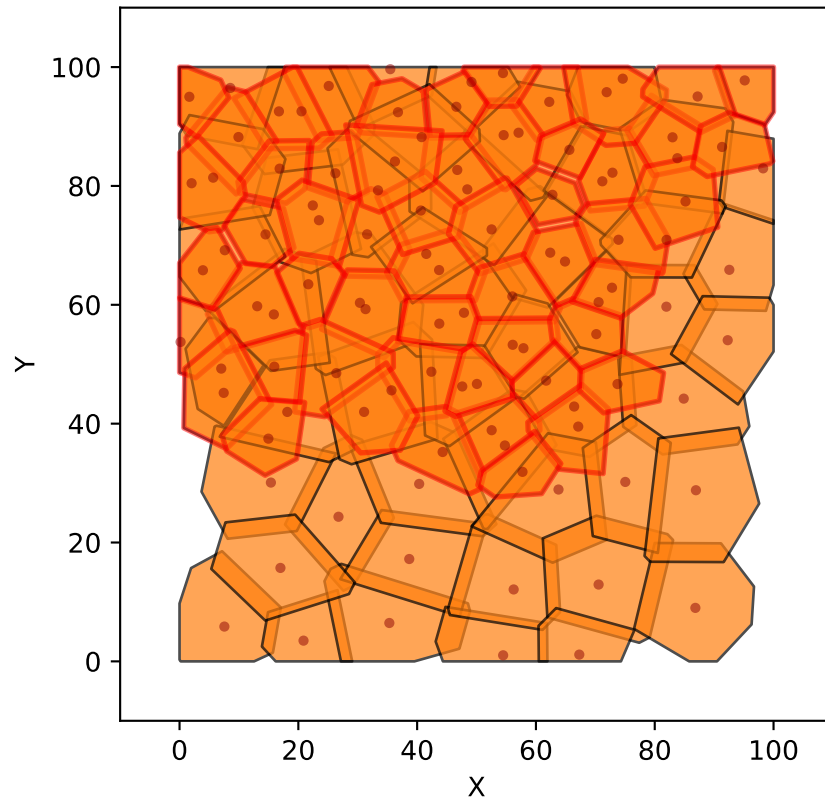
MRF Run Summary	
ARI:	0.0603
errors:	53 / 148
violations:	173 → 170
n_changed:	1
n_iters:	2
n_frozen:	138
Parameters:	
lam:	20.0
tau_low:	0.1
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9

moons — GMM + MRF (signed ramp, tau_low=0.3) ARI=0.052 violations 75 -> 75 changed=0 iters=1

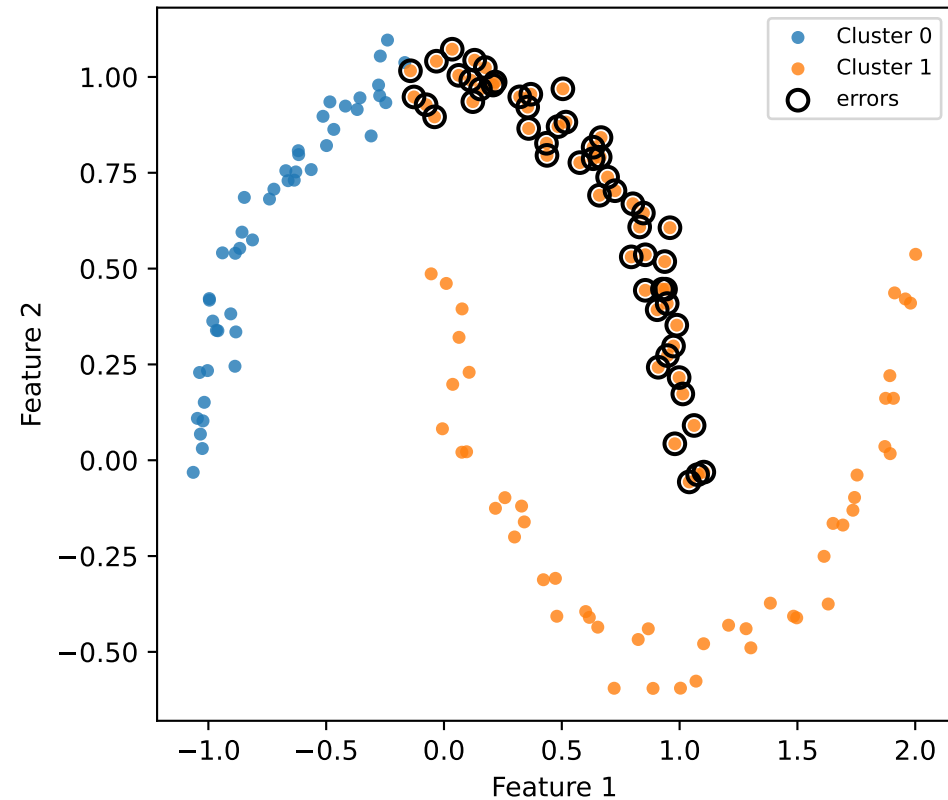
Cluster 0
45 cells



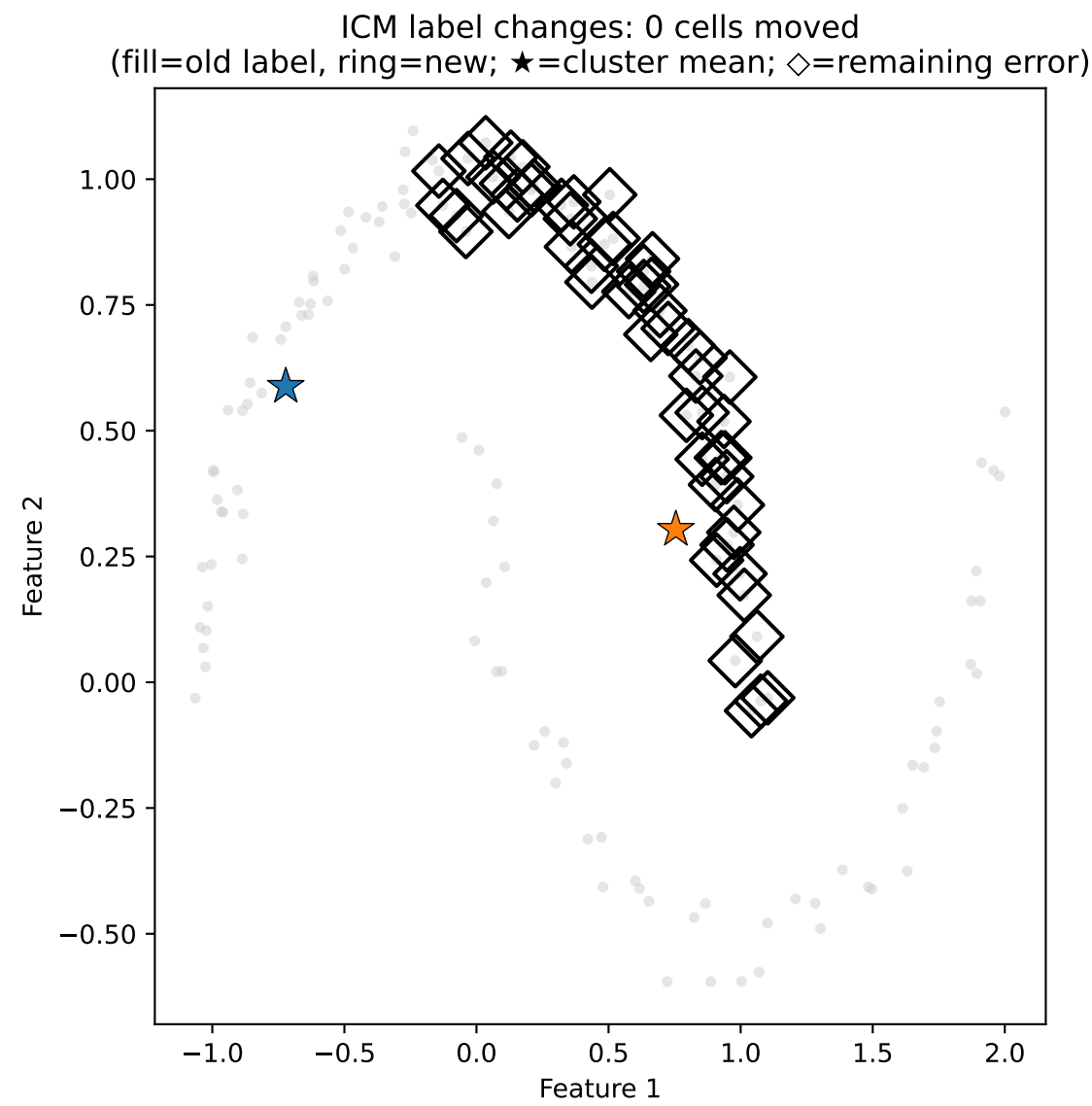
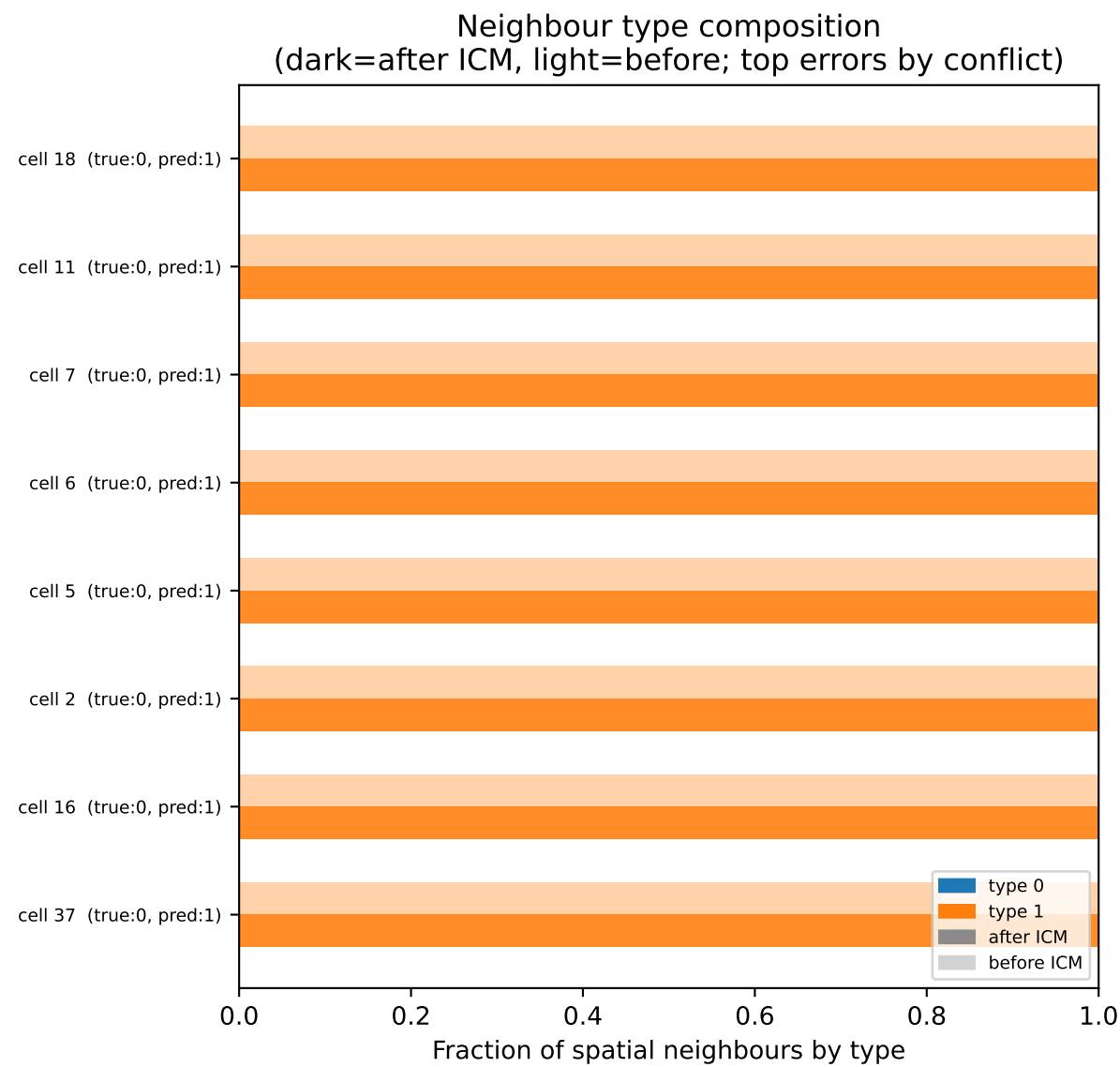
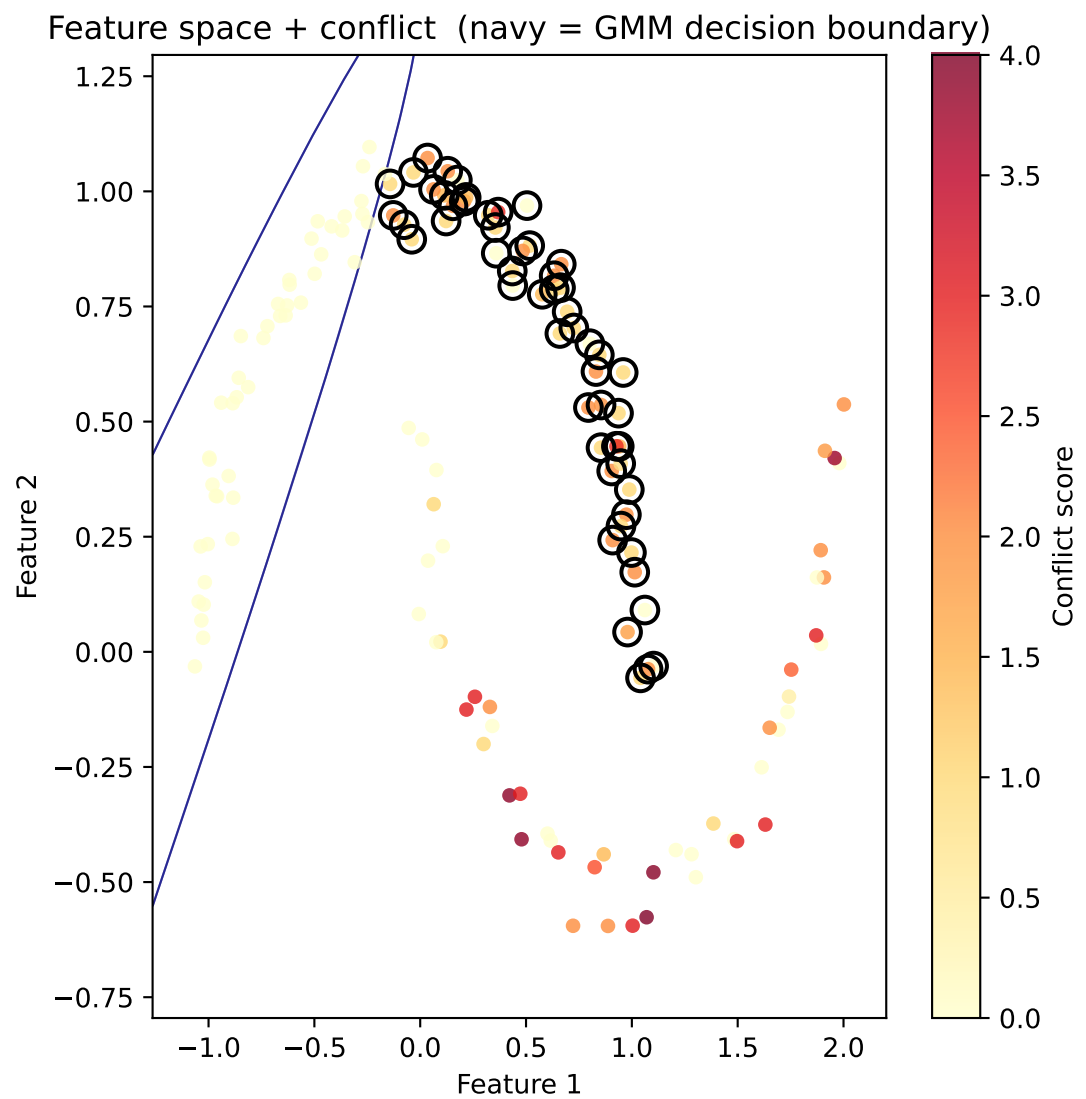
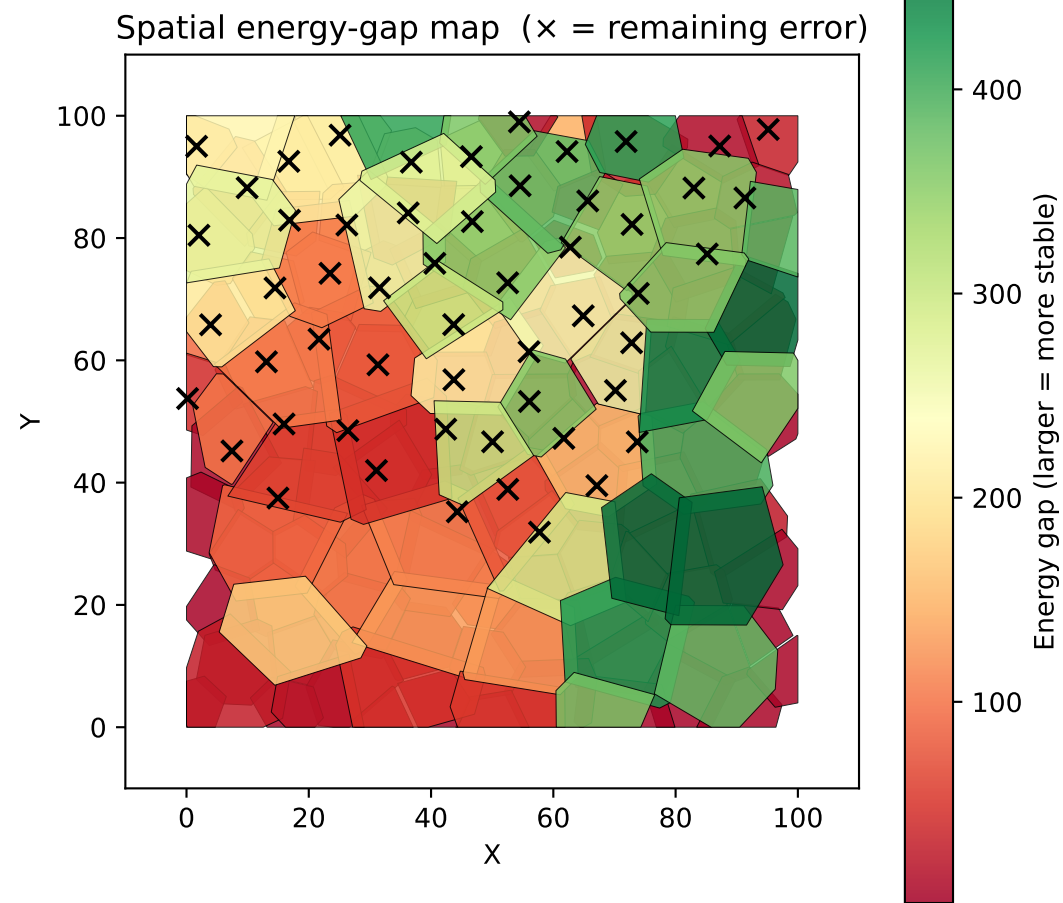
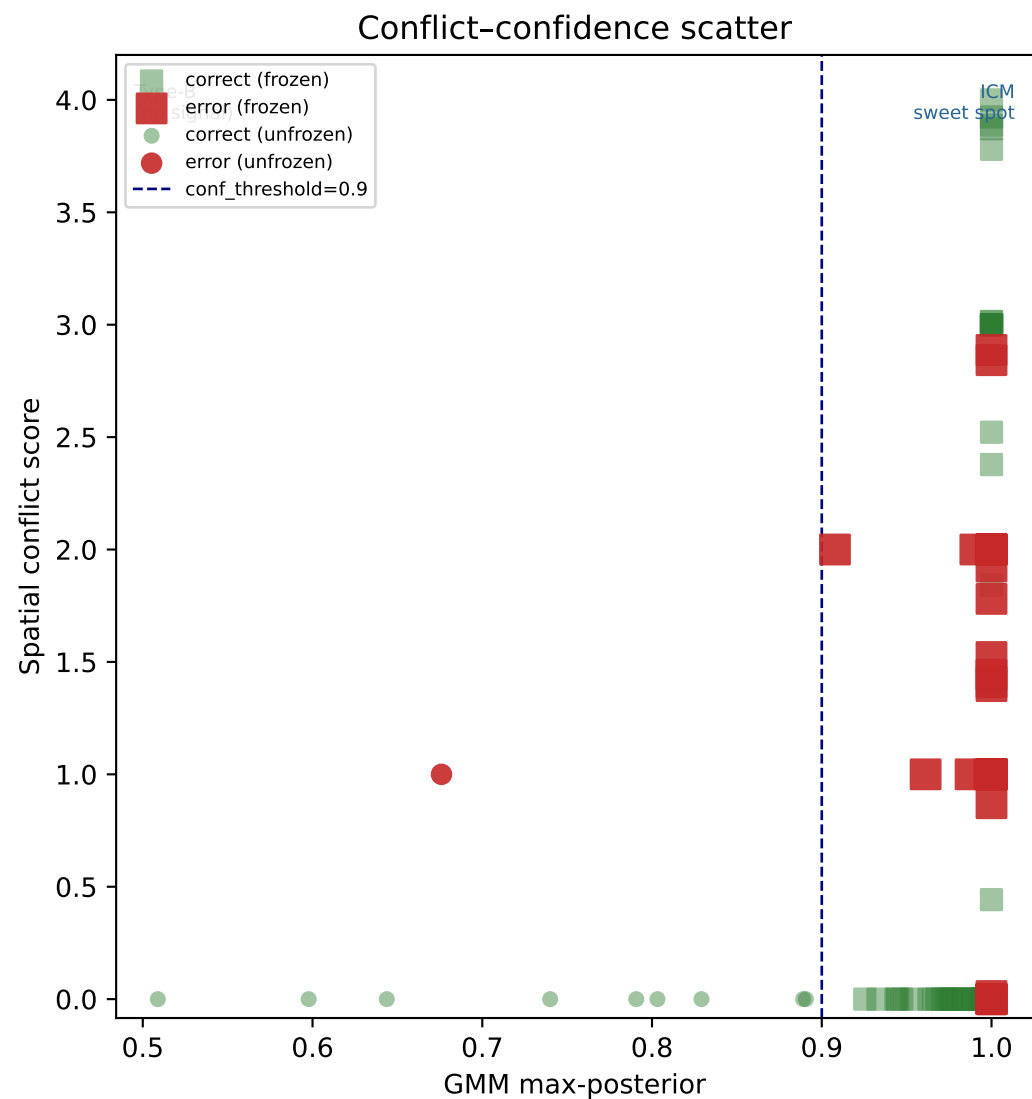
Cluster 1
103 cells (54 err)



Feature space



moons — MRF diagnostics (signed ramp)

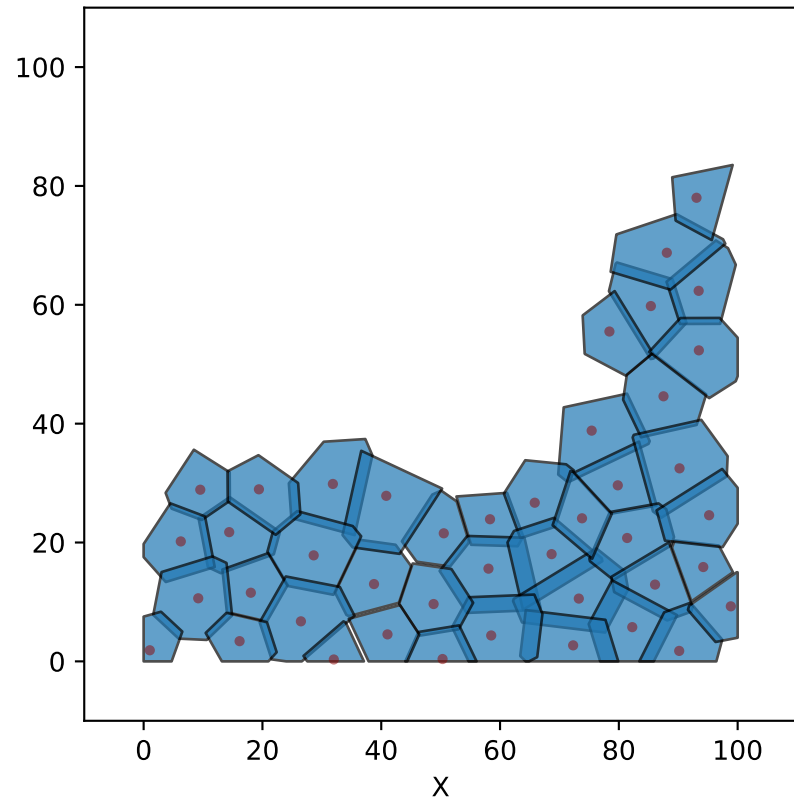


Run summary

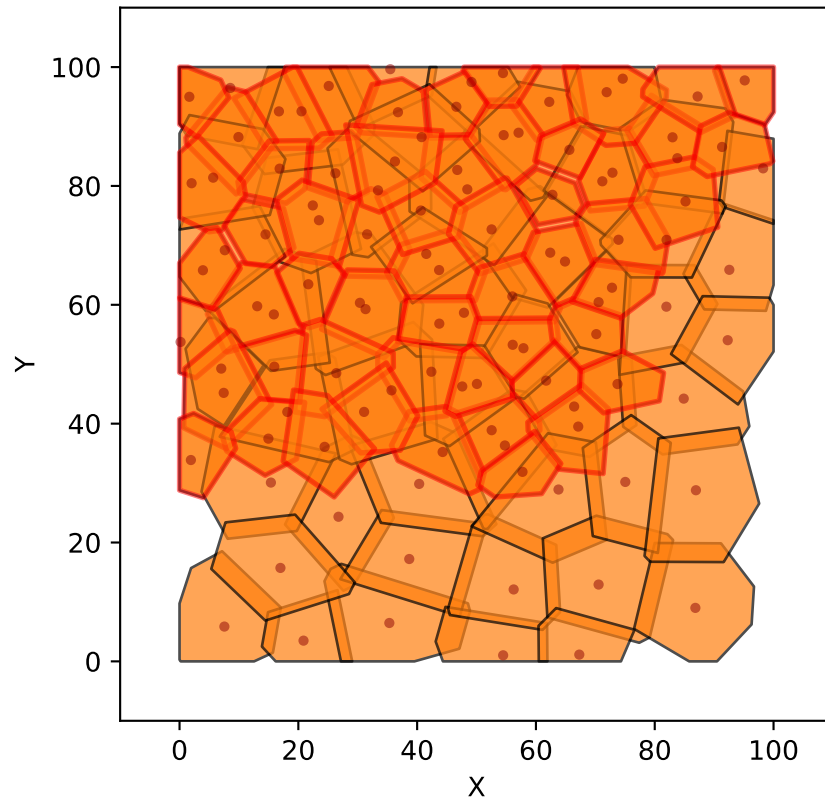
MRF Run Summary	
ARI:	0.0516
errors:	54 / 148
violations:	75 → 75
n_changed:	0
n_iters:	1
n_frozen:	138
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9

moons — GMM + MRF (signed log-ramp $\alpha=10$, $\tau_{\text{low}}=0.3$) ARI=0.035 violations 75 -> 78 changed=2 iters=2

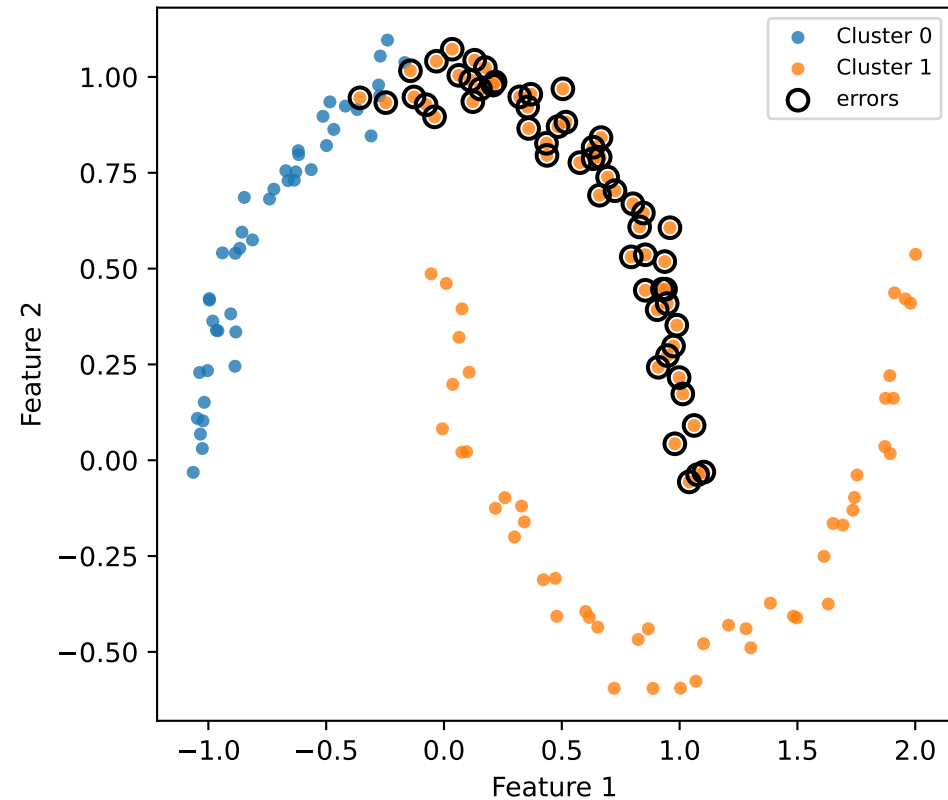
Cluster 0
43 cells

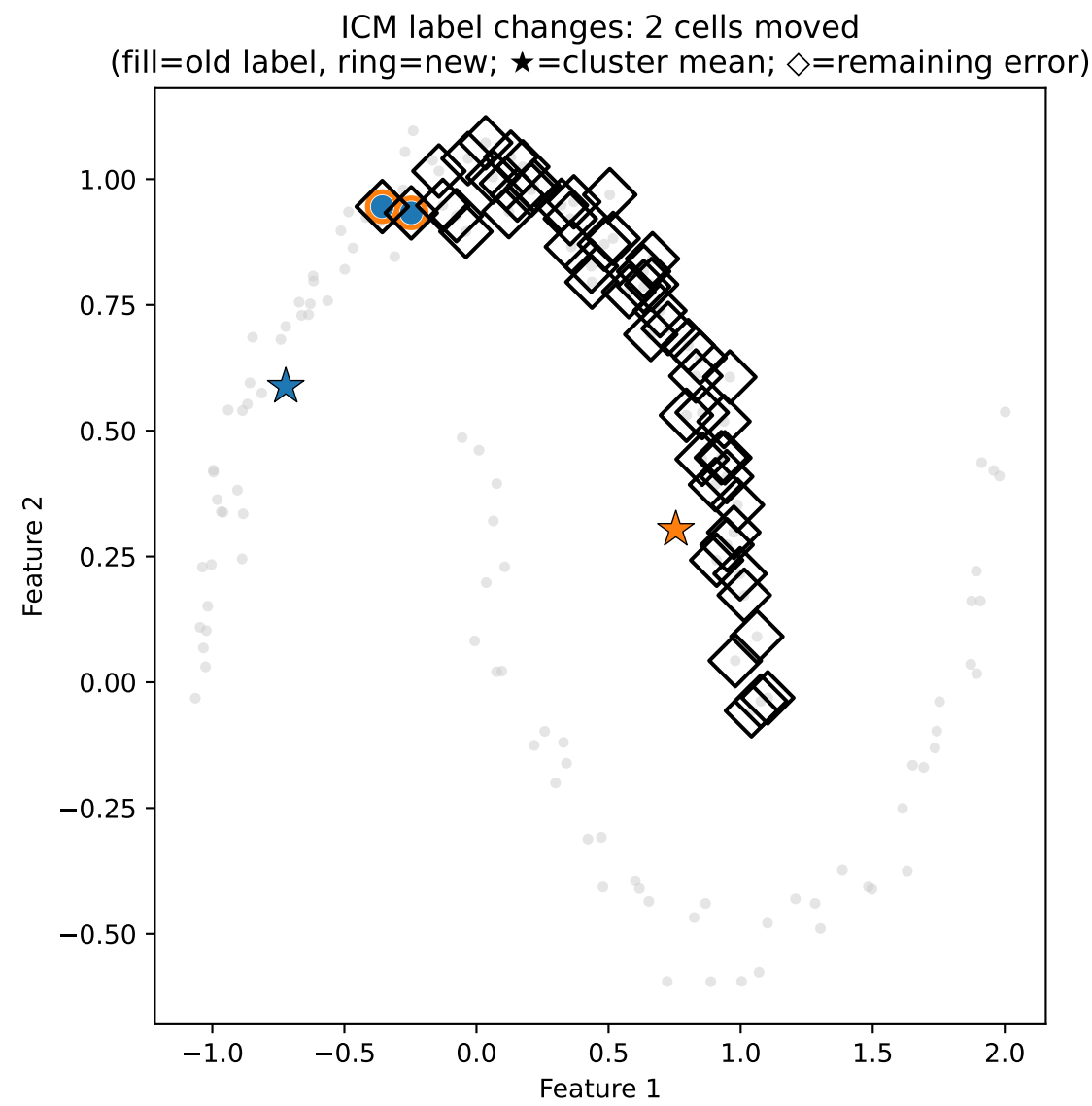
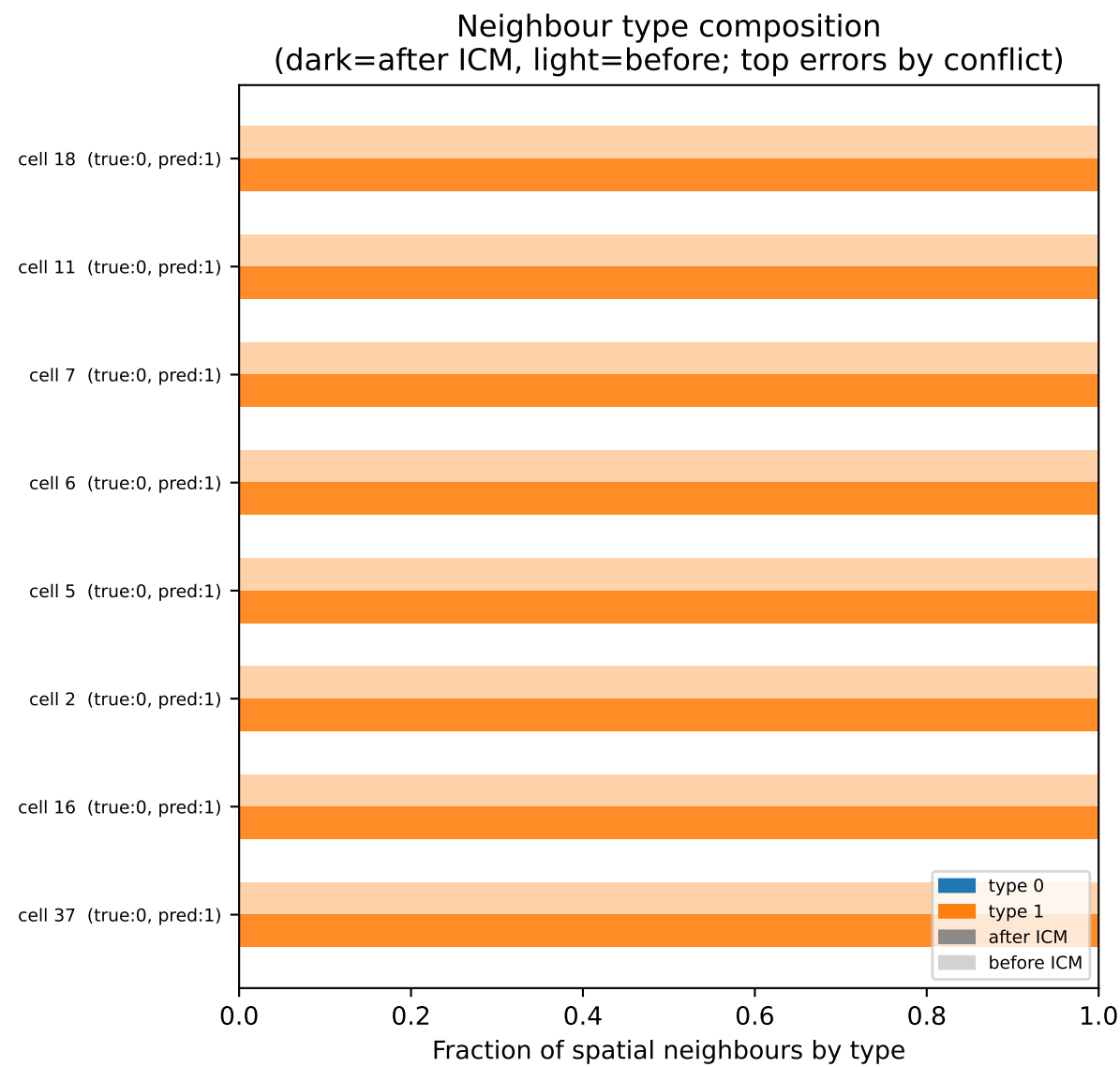
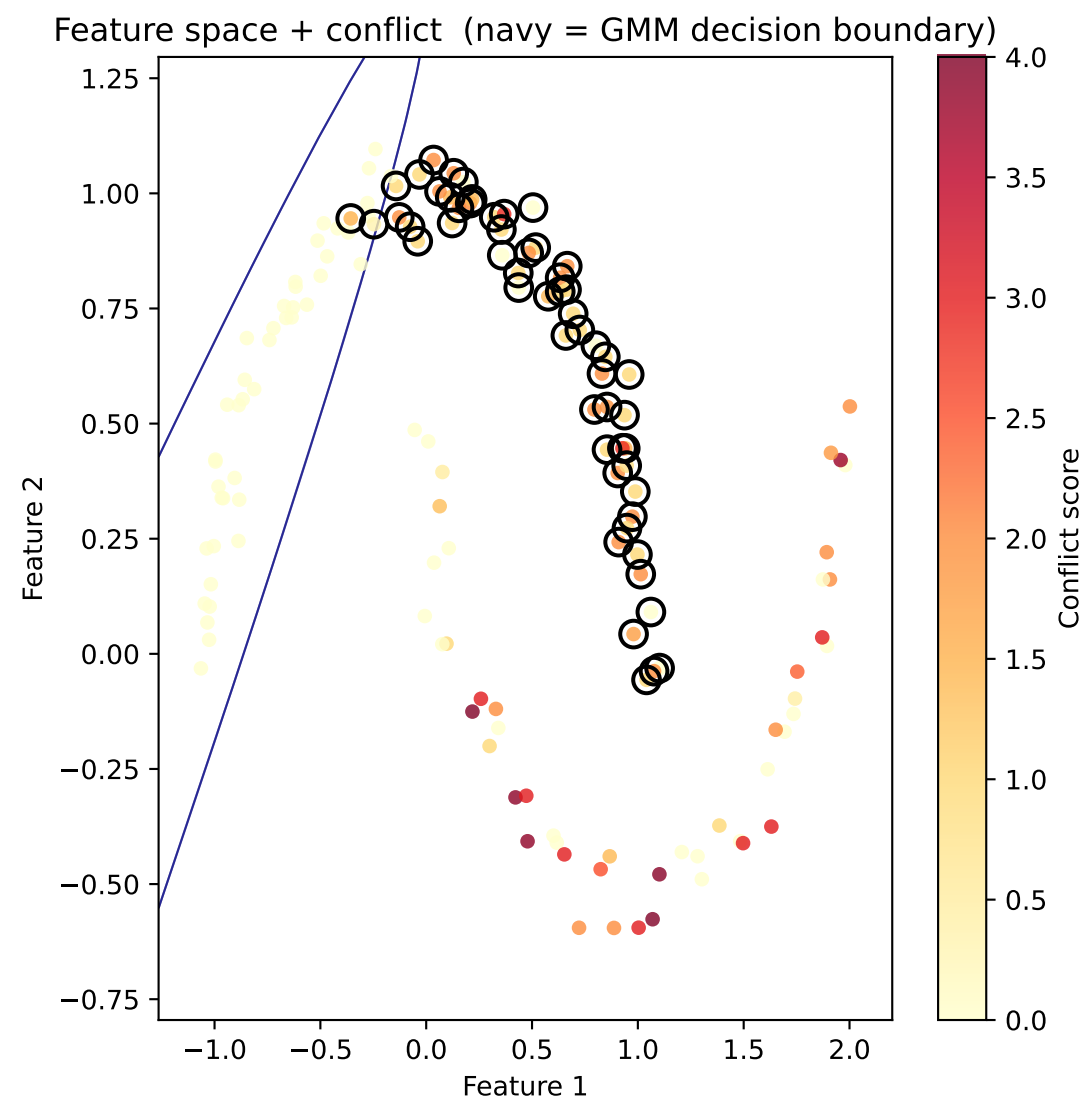
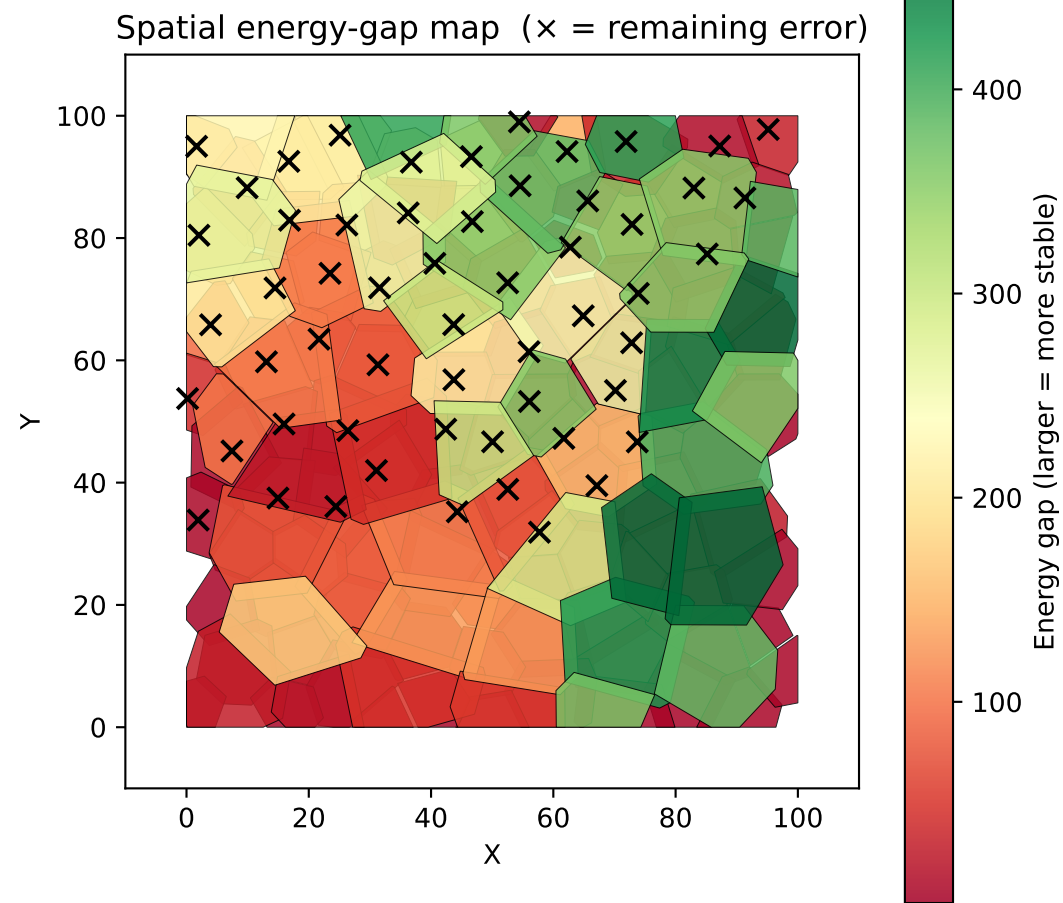
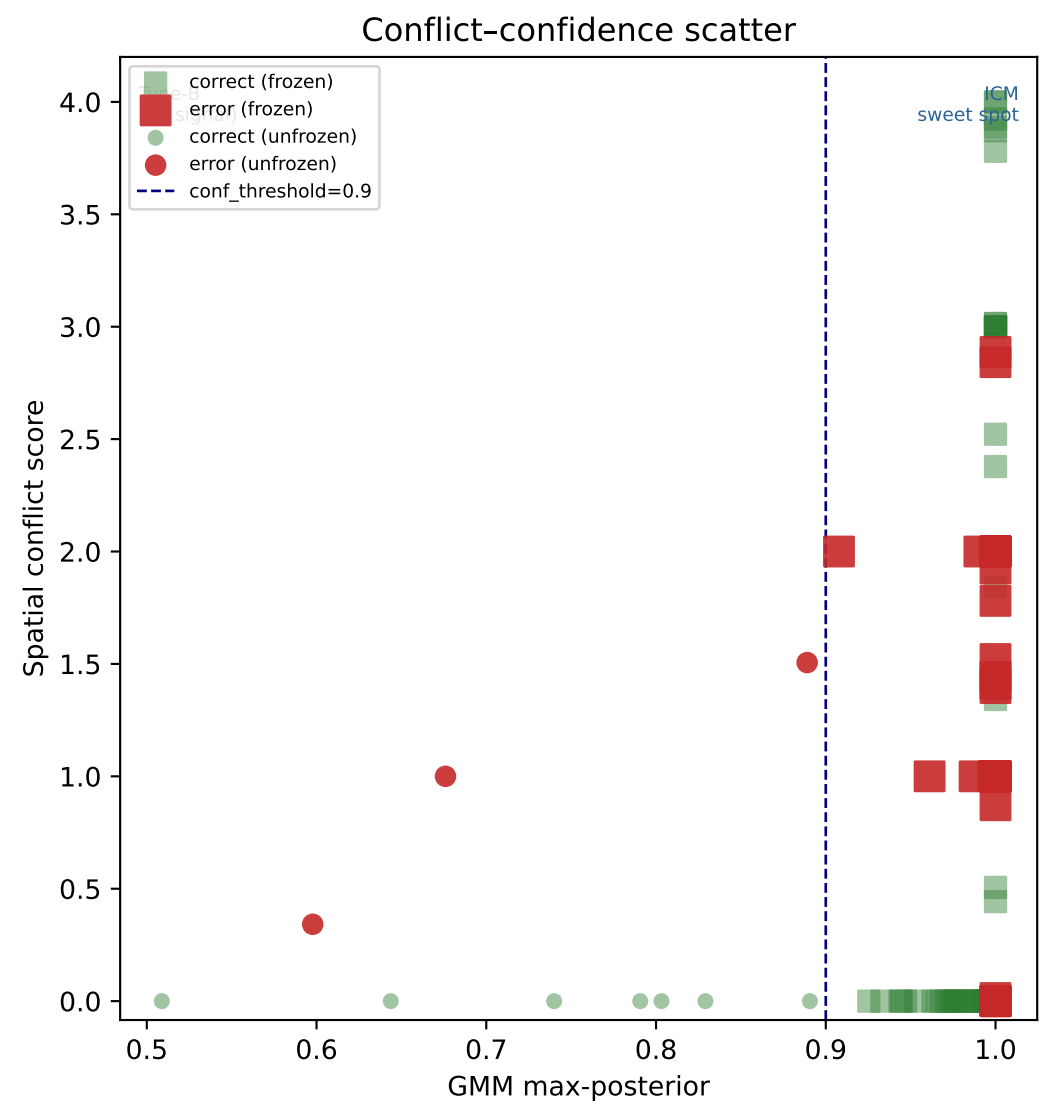


Cluster 1
105 cells (56 err)



Feature space



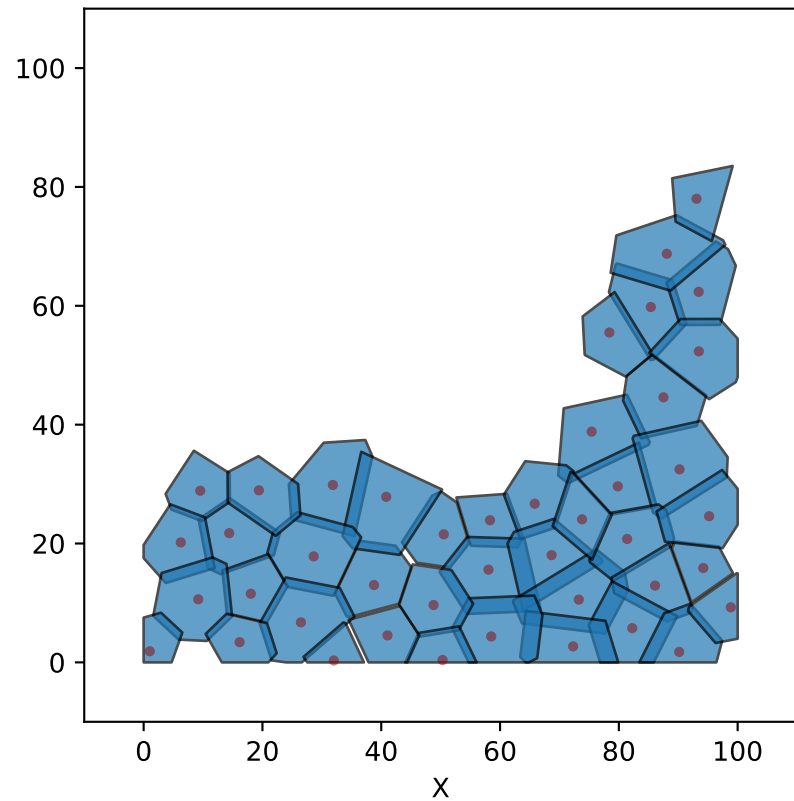


Run summary

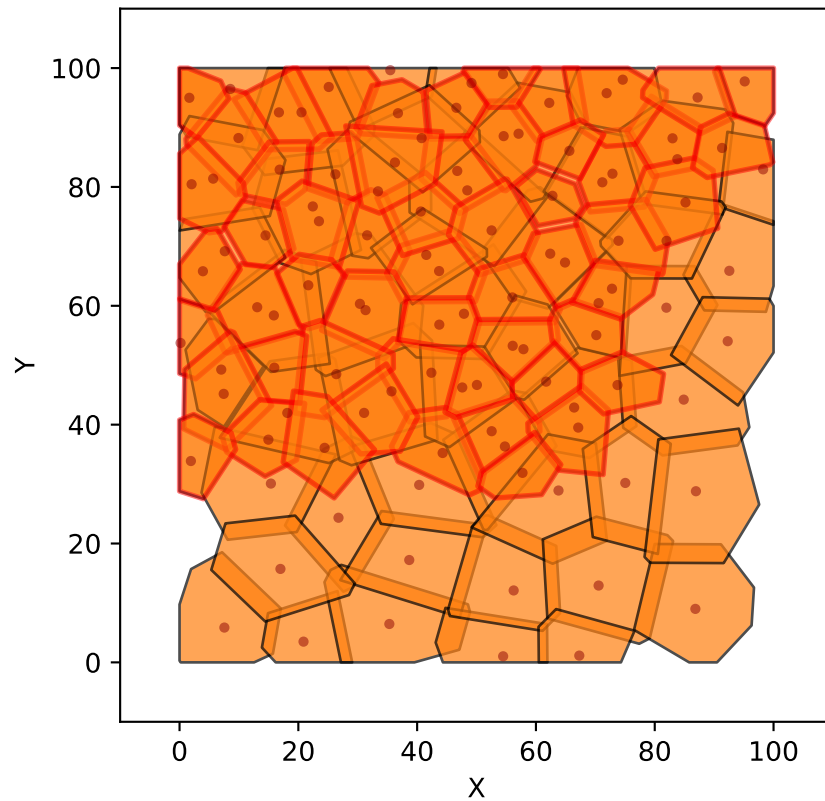
MRF Run Summary	
ARI:	0.0350
errors:	56 / 148
violations:	75 → 78
n_changed:	2
n_iters:	2
n_frozen:	138
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9

moons — GMM + MRF (signed log-ramp a=10 + H1 th=0.85) ARI=0.035 violations 75 -> 78 changed=2 iters=2 force_unfrozen=11

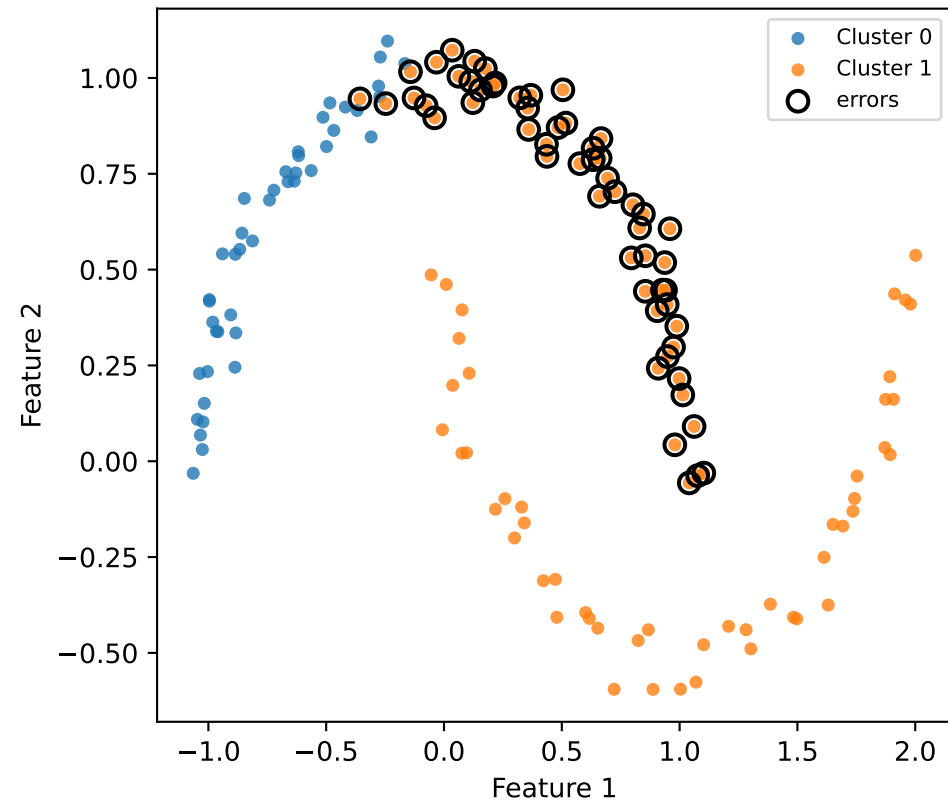
Cluster 0
43 cells

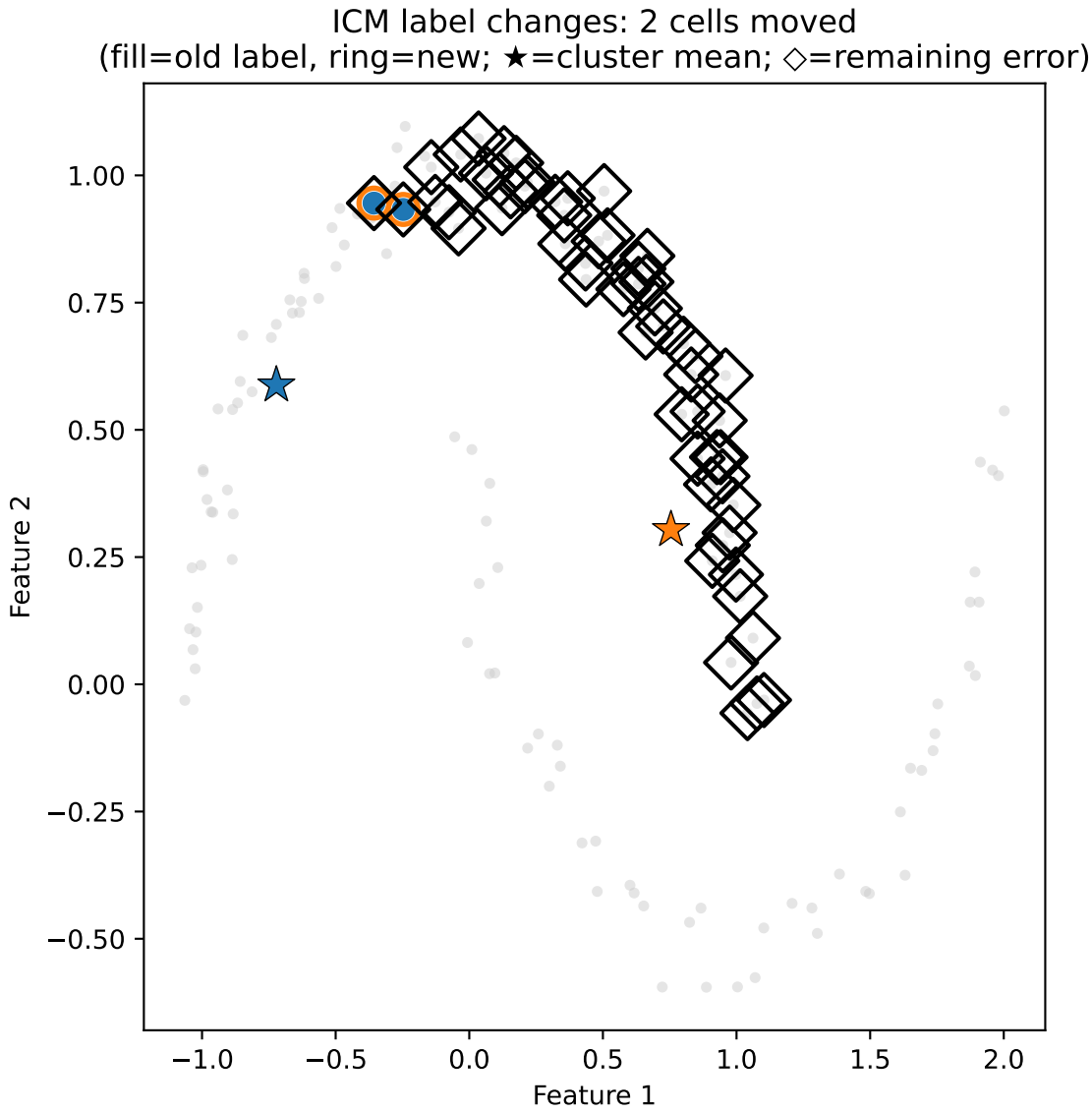
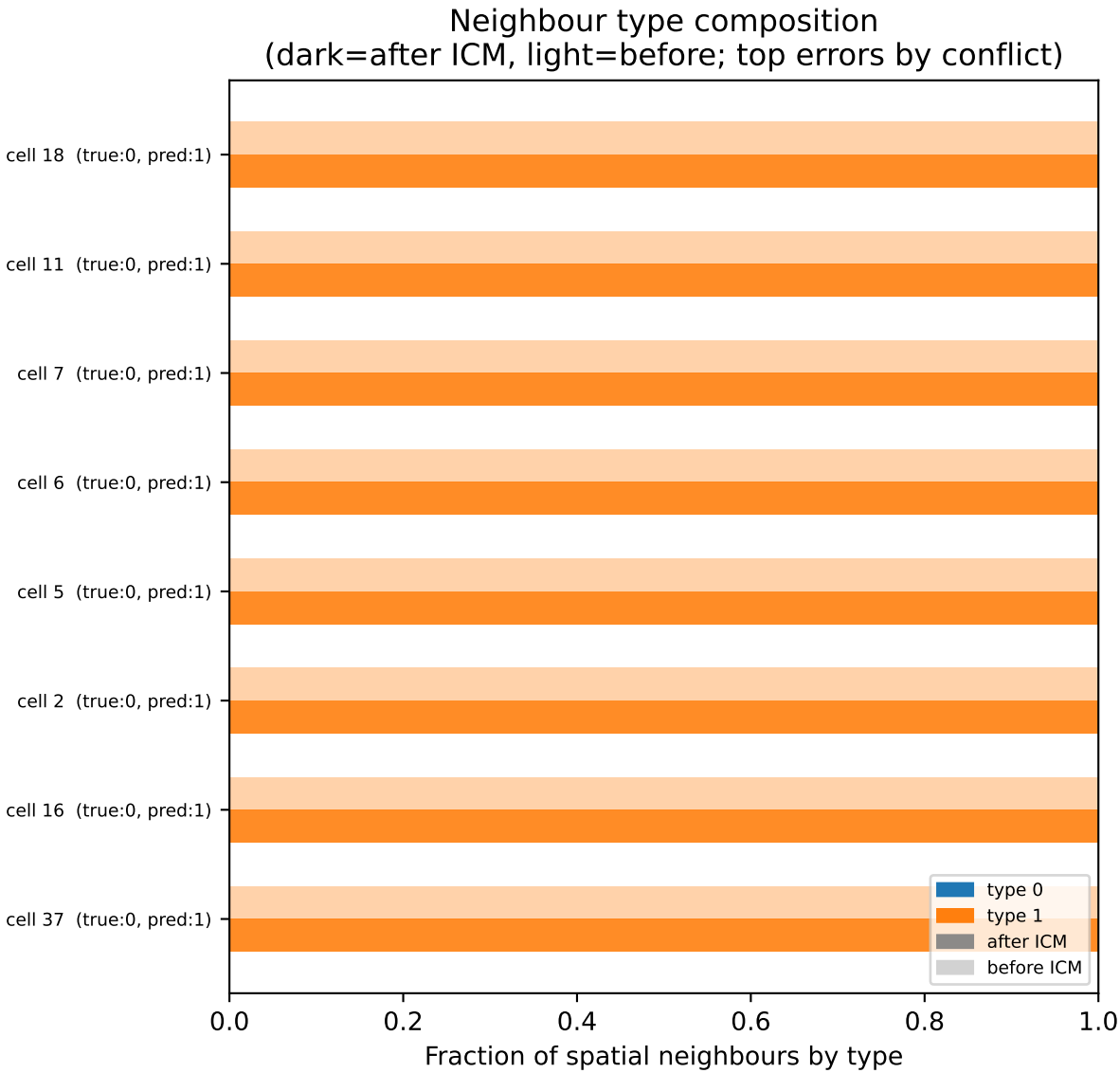
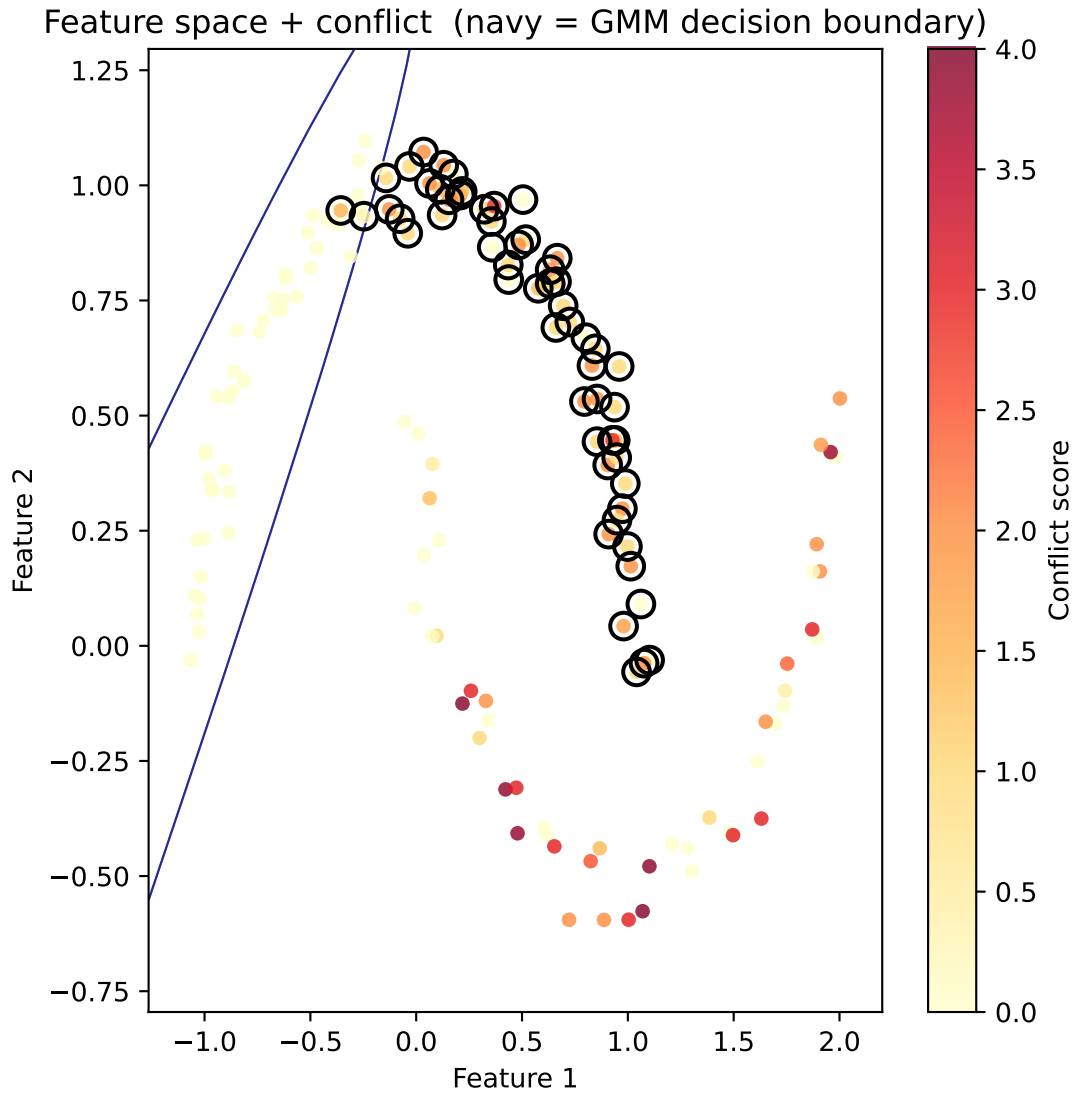
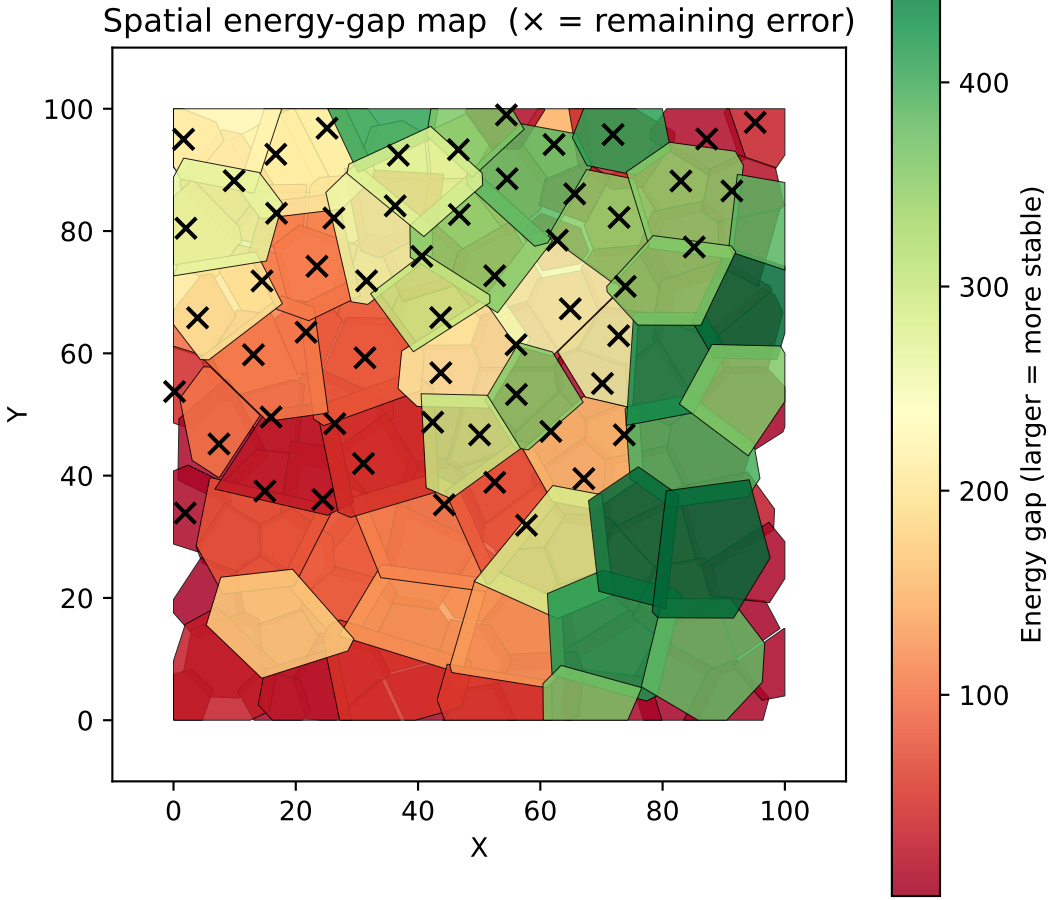
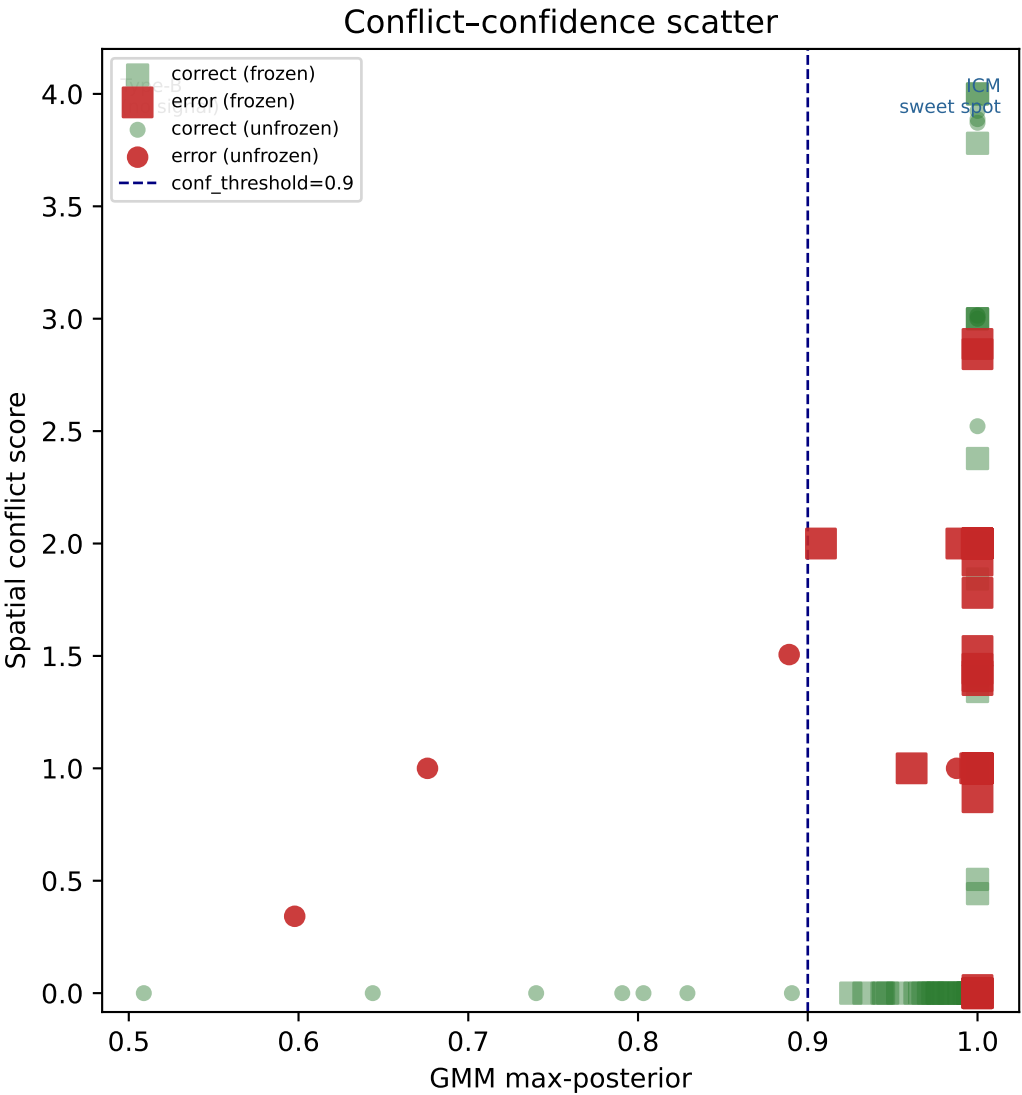


Cluster 1
105 cells (56 err)



Feature space





Run summary

MRF Run Summary	
ARI:	0.0350
errors:	56 / 148
violations:	75 → 78
n_changed:	2
n_iters:	2
n_frozen:	127
Parameters:	
lam:	20.0
tau_low:	0.3
tau_high:	0.4
radius:	15.038613891552956
conf_thr:	0.9